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Induced fermionic currents in de Sitter spacetime in the presence of a compactified cosmic string

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Abstract

We investigate the vacuum fermionic currents in the geometry of a compactified cosmic string on the background of de Sitter (dS) spacetime. The currents are induced by magnetic fluxes running along the cosmic string and enclosed by the compact dimension. We show that the vacuum charge and the radial component of the current density vanish. By using the Abel–Plana summation formula, the azimuthal and axial currents are explicitly decomposed into two parts: the first one corresponds to the geometry of a straight cosmic string, and the second one is induced by the compactification of the string along its axis. For the axial current the first part vanishes and the corresponding topological part is an even periodic function of the magnetic flux along the string axis and an odd periodic function of the flux enclosed by the compact dimension with the periods equal to the flux quantum. The azimuthal current density is an odd periodic function of the flux along the string axis and an even periodic function of the flux enclosed by the compact dimension with the same period. Depending on the magnetic fluxes, the planar angle deficit can either enhance or reduce the azimuthal and axial currents. The influence of the background gravitational field on the vacuum currents is crucial at distances from the string larger than the dS curvature radius. In particular, for the geometry of a straight cosmic string and for a massive fermionic field, we show that the decay of the azimuthal current density is damping oscillatory with the amplitude inversely proportional to the fourth power of the distance from the string. This behavior is in clear contrast with the case of the string in Minkowski bulk, where the current density is exponentially suppressed at large distances.

Keywords: cosmic string, de Sitter space, vacuum current

(Some figures may appear in colour only in the online journal)

1. Introduction

It is well known that geometrical and topological effects play a central role in a large number of physical problems. They have important implications on all scales, from subnuclear to cosmological. In particular, in quantum field theory, the properties of the vacuum crucially depend on the both the geometry and topology of the background spacetime. In the present paper we consider the combined effects of the geometry and topology on the vacuum current densities induced by magnetic flux tubes. As a background geometry we consider de Sitter (dS) spacetime and the topological effects are induced by two types of source. The first one will correspond to a planar angle deficit due to the presence of a cosmic string and the second one comes from the compactification of the spatial dimension along the cosmic string.

Cosmic strings are among the most important types of topological defects that may have been formed by the phase transitions in the early Universe [2]. Though the recent observations of the cosmic microwave background radiation have ruled them out as the primary source for primordial density perturbations, the cosmic strings give rise to a number of interesting physical effects such as the doubling images of distant objects or even gravitational lensing, the emission of gravitational waves and the generation of high-energy cosmic rays (see, for instance, [3]). Recent developments on the formation of topological defects in superstring theories have led to a renewed interest in cosmic (super)strings. In particular, a variant of their formation mechanism has been proposed in the framework of brane inflation [4]. String-like defects also appear in a number of condensed matter systems, including liquid crystals and graphene-made structures.

In the simplest theoretical model, the cosmic string is described by a planar angle deficit with the background geometry being locally flat except on the top of the string where it has a delta shaped curvature tensor. The corresponding nontrivial topology induces nonzero vacuum expectation values (VEVs) for physical observables. Specifically, the VEV of the energy–momentum tensor associated with various fields has been developed by many authors [5–27]. Moreover, considering a magnetic flux running along the strings, there appear additional contributions to the corresponding vacuum polarization effects for charged fields [9, 28–32]. The presence of a magnetic flux induces also vacuum current densities. This phenomenon was analyzed for massless [33] and massive [34] scalar fields. It has been shown that an azimuthal vacuum current appears if the ratio of the magnetic flux by the quantum one has a nonzero fractional part. The analysis of the induced fermionic currents in higher-dimensional cosmic string spacetime in the presence of a magnetic flux have been developed in [35]. The fermionic current induced by a magnetic flux in (2+1)-dimensional conical spacetime and in the presence of a circular boundary has also been analyzed [36].

In general, the analysis of quantum effects for matter fields in a cosmic string spacetime, consider this defect in a flat background geometry. For a cosmic string in a curved background, quantum effects associated with a scalar field have been discussed in [37] for special values of the planar angle deficit. The vacuum polarization in Schwarzschild space-time threaded by an infinite straight cosmic string is investigated in [38]. In recent publications we have investigated the vacuum polarization effects for massive scalar [39] and fermionic [40] fields, induced by a cosmic string in dS spacetime. Similar analysis for vacuum polarization effects, induced by a cosmic string in anti-dS spacetime, have been developed in [41] and [42] for massive scalar and fermionic fields, respectively.

The choice of dS spacetime as the background geometry in the present paper is motivated by several reasons. First of all, this spacetime is a maximally symmetric solution of the Einstein equation in the presence of a positive cosmological constant and, as a consequence of high degree of symmetry, a large number of physical problems are exactly solvable on its background. As it will be shown below, this is the case for the problem under consideration. The importance of dS spacetime as a gravitational background has essentially increased after the appearance of the inflationary scenario for the expansion of the Universe at early stages. Most versions of this scenario assume a period of quasiexponential expansion in which the geometry of the Universe is approximated by a portion of dS spacetime. This gives a natural solution to a number of problems in standard cosmology. In addition, the quantum fluctuations in the inflaton field during the inflationary epoch generate inhomogeneities that are seeds for the formation of the large scale cosmic structures. More recently, astronomical observations of high-redshift supernovae, Galaxy clusters, and the cosmic microwave background have indicated that at present the Universe is accelerating and can be well approximated by the Friedmann–Robertson–Walker cosmological model with the energy dominated by a positive cosmological constant-type source. If the Universe is going to accelerate forever, this model will lead asymptotically to a dS spacetime as a future attractor for the dynamics of the Universe.

The second type of the topological effects we shall consider here is induced by the compactification of the spatial dimension along the cosmic string axis. The compact spatial dimensions are an inherent feature of most high-energy theories of fundamental physics, including supergravity and superstring theories. An interesting application of the field theoretical models with compact dimensions recently appeared in nanophysics. The long-wavelength description of the electronic states in graphene can be formulated in terms of the Dirac-like theory in three-dimensional spacetime with the Fermi velocity playing the role of speed of light (see, e.g., [43]). In graphene-made structures, like cylindrical and toroidal carbon nanotubes, the background geometry for the corresponding field theory contains one or two compact dimensions. In quantum field theory, the periodicity conditions imposed on the field operator along compact dimensions modify the spectrum for the normal modes and as a result of this the VEVs of physical observables are changed. Recently the analysis of the induced fermionic current and the VEV of the energy–momentum tensor in a compactified cosmic string spacetime in the presence of magnetic flux running along the string, have been developed in [44, 45]. The VEV of the fermionic current in spacetimes with an arbitrary number of toroidally compactified spatial dimensions and in the presence of a constant gauge has been investigated in [46]. Furthermore, the combined effects of topology and the gravitational field on the VEVs of the current density for charged scalar and fermionic fields in the background of dS spacetime with an arbitrary number of toroidally compactified spatial dimensions is considered in [47]. The finite temperature effects on the current densities for scalar and fermionic fields in topologically nontrivial spaces have been discussed in [48, 49].

The present paper is organized as follows. In section 2 we describe the background geometry and construct the complete set of normalized positive- and negative-energy fermionic mode functions obeying a quasiperiodic boundary condition with an arbitrary phase along the string axis. In addition, we assume the presence of a constant gauge field. In section 3, by using the mode-summation method, we first show that the VEVs for the charge density and the radial current vanish. Then we evaluate the renormalized VEV of the azimuthal current density induced by a magnetic flux running along the string axis. It is decomposed into two parts: the first one corresponds to the geometry of a cosmic string in dS spacetime without compactification and the second one is induced by the compactification of the spatial dimension parallel to the string. The VEV of the axial current density is investigated in section 4. This VEV is a purely topological effect induced by the compactification and vanishes in the geometry of a straight cosmic string. The most relevant conclusions of the paper are summarized in section 6. Throughout the paper we use the units with $G = \hbar = c = 1$.

2. Geometry of the problem and the fermionic modes

The main objective of this section is to present the geometry of the spacetime, where we develop our analysis and also to obtain the complete set of solutions of Dirac equation in this background. So we first write the line element, in cylindrical coordinates, corresponding to a cosmic string along the z-axis in dS spacetime

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = dt^2 - e^{2t/\alpha}\left(dr^2 + r^2 d\phi^2 + dz^2\right), \tag{2.1}$$

where $r \geqslant 0$, $t \in (-\infty, +\infty)$ and $0 \leqslant \phi \leqslant \phi_0$, being $\phi_0 = 2\pi/q$. The parameter q, bigger than unity, codifies the presence of the cosmic string and the parameter α is related to the cosmological constant Λ and the scalar curvature R by the expressions $\Lambda = 3\alpha^{-2}$ and $R = 12\alpha^{-2}$. Additionally we shall assume that the direction along the z-axis is compactified to a circle with the length L: $0 \leqslant z \leqslant L$. The physical consequences of modifying the topology of dS spacetime have been discussed in [50]. It was argued that for dS spacetime which may emerge from string theory the most natural topology is that with toroidally compact dimensions. The latter arise very naturally in quantum cosmology. The models with compact spatial dimensions may play an important role by providing proper initial conditions for inflation [51]. In these models the probability of inflation is not exponentially suppressed.

The dynamics of a massive spinor field in curved spacetime in the presence of a four-vector potential, A_μ, is governed by the Dirac equation

$$i\gamma^\mu\left(\nabla_\mu + ieA_\mu\right)\psi - m\psi = 0, \quad \nabla_\mu = \partial_\mu + \Gamma_\mu, \tag{2.2}$$

where Γ_μ is the spin connection. In what follows, we will take the Dirac matrices according to

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^l = e^{-t/\alpha}\begin{pmatrix} 0 & \rho^l \\ -\rho^l & 0 \end{pmatrix}, \tag{2.3}$$

with the 2×2 matrices

$$\rho^1 = \begin{pmatrix} 0 & e^{-iq\phi} \\ e^{iq\phi} & 0 \end{pmatrix}, \quad \rho^2 = \frac{1}{ir}\begin{pmatrix} 0 & e^{-iq\phi} \\ -e^{iq\phi} & 0 \end{pmatrix}, \quad \rho^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.4}$$

We assume that along the compact z-dimension the field obeys the quasiperiodicity condition

$$\psi(t, r, \phi, z + L) = e^{2\pi i\beta}\psi(t, r, \phi, z), \tag{2.5}$$

where β is a constant defined in the interval [0, 1]. The special cases $\beta = 0$ and $\beta = 1/2$ correspond to the periodic and antiperiodic boundary conditions (untwisted and twisted fields, respectively). For the rotation around the z-axis we shall use the periodic boundary condition

$$\psi\left(t, r, \phi + \phi_0, z\right) = \psi(t, r, \phi, z). \tag{2.6}$$

For a constant vector potential, A_μ, the latter may be excluded from the field equation (2.2) by the gauge transformation $A'_\mu = A_\mu + \partial_\mu \Lambda$, $\psi' = \mathrm{e}^{-\mathrm{i}e\Lambda}\psi$, with $\Lambda = -A_\mu x^\mu$. The new wave function obeys the equation $(\mathrm{i}\gamma^\mu \nabla_\mu - m)\psi' = 0$ and the periodicity conditions

$$\psi'\left(t, r, \phi + \phi_0, z\right) = \mathrm{e}^{2\pi\mathrm{i}a}\psi'(t, r, \phi, z), \tag{2.7}$$

$$\psi'(t, r, \phi, z + L) = \mathrm{e}^{2\pi\mathrm{i}\tilde{\beta}}\psi'(t, r, \phi, z), \tag{2.8}$$

with the notations

$$a = \mathrm{e}A_2/q, \;\; \tilde{\beta} = \beta + \mathrm{e}A_3 L/2\pi. \tag{2.9}$$

Note that the physical components A_ϕ and A_z of the vector potential are related to the covariant components A_2 and A_3 by $A_\phi = -A_2/r$ and $A_z = -A_3$. The parameters in the phases of the periodicity conditions can be expressed in terms of the magnetic flux along the string axis, $\Phi_2 = -A_2\phi_0$, and flux enclosed by the z -axis, $\Phi_3 = -A_3 L$, by the formulas

$$a = -\Phi_2/\Phi_0, \;\; \tilde{\beta} = \beta - \Phi_3/\Phi_0, \tag{2.10}$$

with $\Phi_0 = 2\pi/e$ being the flux quantum. In what follows we will work in terms of the gauge transformed field ψ' omitting the prime. The current density is invariant under the gauge transformation.

Our main interest in this paper is the evaluation of the VEV of the fermionic current density, $j^\mu = e\bar{\psi}\gamma^\mu\psi$. This VEV is expressed in terms of the two-point function $S^{(1)}_{rs}(x, x') = \langle 0|[\psi_r(x), \bar{\psi}_s(x')]|0\rangle$, where r and s are spinor indices and $|0\rangle$ is the vacuum state. For the VEV one has

$$\langle j^\mu(x)\rangle \equiv \langle 0|\, j^\mu(x)|0\rangle = -\frac{e}{2}\lim_{x'\to x}\mathrm{Tr}\left(\gamma^\mu S^{(1)}\left(x, x'\right)\right). \tag{2.11}$$

In quantum field theory on curved backgrounds the choice of the vacuum is not unique (see, for example, [1]). In dS spacetime there exists a one-parameter family of maximally symmetric quantum states. In what follows we will assume that the field is prepared in the dS-invariant Bunch–Davies vacuum state [52]. In the class of dS-invariant quantum states, the Bunch–Davies vacuum is the only one for which the ultraviolet behavior of the two-point functions is the same as in Minkowski spacetime.

Let $\{\psi^{(+)}_\sigma(x), \psi^{(-)}_\sigma(x)\}$ be a complete set of normalized solutions to the Dirac equation specified by the set of quantum numbers σ. Note that the background geometry under consideration is time-dependent and the energy is not conserved. However, we will refer to the solutions $\psi^{(+)}_\sigma(x)$ and $\psi^{(-)}_\sigma(x)$ as the positive- and negative-energy modes in the sense that in the limit $\alpha \to \infty$ they reproduce the positive- and negative-energy fermionic modes in Minkowski spacetime. Expanding the field operator in terms of the complete set of fermionic modes, the following mode-sum formula is obtained for the current density:

$$\langle j^\mu\rangle = \frac{e}{2}\sum_\sigma\left[\bar{\psi}^{(-)}_\sigma(x)\gamma^\mu\psi^{(-)}_\sigma(x) - \bar{\psi}^{(+)}_\sigma(x)\gamma^\mu\psi^{(+)}_\sigma(x)\right]. \tag{2.12}$$

Consequently, in this evaluation we need the fermionic modes for the geometry at hand.

The positive- and negative-energy mode functions obeying the periodicity conditions (2.8) can be found in a way similar to that we have used in [40] for the geometry of a straight cosmic string in dS spacetime in the absence of the magnetic flux. For the Bunch–Davies vacuum state these functions are given by

$$\psi_{\sigma}^{(\pm)}(x) = C_{\sigma}^{(\pm)}\eta^2 e^{iq(j+a)\phi+ikz}\begin{pmatrix} H_{1/2-im\alpha}^{(\lambda_\pm)}(\gamma\eta)J_{\beta_1}(pr)e^{-iq\phi/2} \\ \frac{isp\epsilon_j}{\gamma+sk}H_{1/2-im\alpha}^{(\lambda_\pm)}(\gamma\eta)J_{\beta_2}(pr)e^{iq\phi/2} \\ -isH_{-1/2-im\alpha}^{(\lambda_\pm)}(\gamma\eta)J_{\beta_1}(pr)e^{-iq\phi/2} \\ \frac{p\epsilon_j}{\gamma+sk}H_{-1/2-im\alpha}^{(\lambda_\pm)}(\gamma\eta)J_{\beta_2}(pr)e^{iq\phi/2} \end{pmatrix}, \tag{2.13}$$

where $\lambda_+ = 1$, $\lambda_- = 2$, $j = \pm 1/2, \pm 3/2, \ldots$, $\eta = \alpha e^{-t/\alpha}$, $s = \pm 1$. Moreover, $J_\nu(x)$ and $H_\nu^{(1,2)}(x)$ are the Bessel and Hankel functions, respectively, and $\gamma = \sqrt{k^2 + p^2}$, $0 \leqslant p < \infty$. In (2.13),

$$\beta_1 = q\,|j + a| - \epsilon_j/2,\ \ \beta_2 = q\,|j + a| + \epsilon_j/2, \tag{2.14}$$

with $\epsilon_j = 1$ for $j > -a$ and $\epsilon_j = -1$ for $j < -a$. The quantum number j determines the eigenvalues of the projection of the total momentum along the cosmic string,

$$\hat{J}_3\psi_{\sigma}^{(\pm)}(x) = \left(-i\partial_{q\phi} + i\gamma^{(1)}\gamma^{(2)}/2\right)\psi_{\sigma}^{(\pm)}(x) = (j + a)\psi_{\sigma}^{(\pm)}(x), \tag{2.15}$$

and the quantum number s corresponds to the eigenvalue of

$$\hat{S} = \gamma^{-1}\Sigma^n\hat{p}_n,\ \ \Sigma^n = \begin{pmatrix} \rho^n & 0 \\ 0 & \rho^n \end{pmatrix}, \tag{2.16}$$

where $\hat{p}_n = -i\partial_n + \delta_n^3(q-1)\Sigma^3/2$ with $n = 1, 2, 3$. Note that the momentum is quantized in units $j - \Phi_2/\Phi_0$ (for physical systems with this type of phenomena see [53]).

The mode functions above are specified by the complete set of quantum numbers $\sigma = (p, k, j, s)$. In addition, the functions (2.13) obey the periodicity condition (2.7). From the condition (2.8) we find the eigenvalues for the quantum number k:

$$k = k_l = 2\pi\left(l + \tilde{\beta}\right)/L, \tag{2.17}$$

with $l = 0, \pm 1, \pm 2, \ldots$. By using the Wronskian for the Hankel functions, from the the orthonormalization condition for the mode functions we find

$$\left|C_{\sigma}^{(\pm)}\right|^2 = \frac{qpe^{\pm m\alpha\pi}}{16L\alpha^3}(\gamma + sk). \tag{2.18}$$

Note that, if we write the parameter a, defined in (2.9), in the form

$$a = n_0 + a_0,\quad |a_0| < 1/2, \tag{2.19}$$

where n_0 is an integer number, then, by shifting $j + n_0 \to j$, we can see that the VEVs of physical observables depend on a_0 only.

As it is well known, in Minkowski spacetime, the theory of von Neumann deficiency indices leads to a one-parameter (usually denoted by θ) family of allowed boundary conditions in the background of an Aharonov–Bohm gauge field [54]. Additionally to the regular modes, these boundary conditions, in general, allow normalizable irregular modes. A special case of boundary conditions has been discussed in [55], where the Atiyah–Patodi–Singer type nonlocal boundary condition is imposed at a finite radius, which is then taken to zero. Similar approach, with the MIT bag boundary condition, has been used in [36, 56] for a two-dimensional conical space with a circular boundary. In the geometry under consideration there are no normalizable irregular modes for $|a_0| \leqslant (1 - 1/q)/2$. In the case $|a_0| > (1 - 1/q)/2$, the irregular mode corresponds to $j = -n_0 - \text{sgn}\,(a_0)/2$. For the mode functions (2.13) with this value of the momentum, the boundary condition on the string axis is a special case of one-parameter family of conditions with the parameter $\theta = \pi/2$. Note that with this value and for a massless field both parity and chiral symmetry are conserved [57]. The evaluation of the VEV of the fermionic current for other boundary conditions on the string axis is similar to that described below. The contribution of the regular modes to the VEV is the same for all boundary conditions and the results will differ by the parts related to the irregular modes.

3. Charge, radial and azimuthal currents

Having the complete set of mode functions (2.13), we can evaluate the VEV for the current density by making use of the mode-sum formula (2.12) where the summation is specified by

$$\sum_{\sigma} = \int_0^\infty \mathrm{d}p \sum_{l=-\infty}^{+\infty} \sum_{s=\pm 1} \sum_j, \tag{3.1}$$

with $\sum_j = \sum_{j=\pm 1/2, \pm 3/2, \ldots}$. Of course, the expression in the rhs of (2.12) is divergent and a regularization with the subsequent renormalization is necessary. Various regularization procedures can be used. For the problem at hand, from the point of view of the brevity for the presentation, the most convenient one is the Pauli–Villars gauge-invariant regularization.

In the Pauli–Villars regularization procedure, regulator fields with large masses are introduced. The number of these fields depends on the specific problem. As it will be seen below, for the problem under consideration a single regulator field with the mass m_1 is sufficient. First let us consider the charge density, $\langle j^0 \rangle$. The corresponding regularized VEV can be presented as

$$\begin{aligned}\langle j^0 \rangle_{\mathrm{reg}} = -\frac{eq\eta^4}{4\pi^2 L\alpha^3} \int_0^\infty \mathrm{d}p\, p \sum_{l=-\infty}^{+\infty} \gamma \sum_j \left[J_{\beta_1}^2(pr) + J_{\beta_2}^2(pr) \right] \\ \times \sum_{n=0,1} c_n \sum_{\chi=-,+} \chi \left[\left| K_{1/2+\chi \mathrm{i} m_n \alpha}(\mathrm{i}\gamma\eta) \right|^2 + \left| K_{-1/2+\chi \mathrm{i} m_n \alpha}(\mathrm{i}\gamma\eta) \right|^2 \right],\end{aligned} \tag{3.2}$$

where $c_0 = 1$, $m_0 = m$, and $K_\nu(x)$ is the MacDonald function. By using the asymptotic expression of the function $K_\nu(\mathrm{i}x)$ for large values x, we can see that with the choice $c_1 = -1$ the leading term in the expansion over $1/\gamma$ is canceled by the regulator field. By increasing the number of regulator fields the subleading terms can be canceled as well by an appropriate choice of the coefficients. However, for the case under consideration a single additional field is sufficient. Now, by taking into account that $K_{-\nu}(x) = K_\nu(x)$, from (3.2) we conclude that the charge density vanishes. For the VEV of the radial current density, $\langle j^1 \rangle$, substituting the fermionic mode functions from (2.13) into the corresponding mode-sum, it can be seen that all terms in the regularized VEV cancel and the resulting radial current is also zero.

Now we turn to the azimuthal current which is given by the expression (2.12) with $\mu = 2$. Substituting (2.13), we can see that the positive- and negative-energy modes give the same contribution. After the summation over s, the regularized VEV of the azimuthal current is presented in the form

$$\langle j^2\rangle_{\rm reg} = -\frac{2eq\eta^5}{\pi^2 L\alpha^4 r}\int_0^\infty \mathrm{d}p\, p^2 \sum_j \epsilon_j J_{\beta_1}(pr)J_{\beta_2}(pr) \times \sum_{n=0,1} c_n \sum_{l=-\infty}^{+\infty} \mathrm{Re}\left[K_{1/2+\mathrm{i}m_n\alpha}(\mathrm{i}\gamma\eta)K_{1/2+\mathrm{i}m_n\alpha}(-\mathrm{i}\gamma\eta)\right]. \tag{3.3}$$

It can be seen that the same choice of c_1 as in (3.2) cancels the leading term in the asymptotic expansion and makes the expression (3.3) finite.

The induced VEV of the current density is an Aharonov–Bohm type quantum phenomenon and arises as a result of the Dirac sea distortion by its interaction with the magnetic flux. The physical origin of the current is the flux sensitivity of the mode functions for a quantum field. When the magnetic flux along the cosmic string is a multiple of the flux quantum (i.e. a is an integer number), the series in (3.3) takes the form $\sum_j \mathrm{sgn}(j)J_{q|j|-1/2}(pr)J_{q|j|+1/2}(pr)$ and it vanishes due to the cancelation of the contribution from the modes with opposite signs of j. In the presence of a fractional magnetic flux ($a_0 \neq 0$) this symmetry between the modes with opposite signs of j is broken and there is no cancelation between the contributions from the modes with $j = |j|$ and $j = -|j|$. The total vacuum current is the sum over the individual contributions from each mode and, as a result of this, a nonzero net current density appears.

In order to separate explicitly the part in the current induced by the compactification, for the summation over l in (3.3), we apply the Abel–Plana formula in the form [58, 59]

$$\frac{2\pi}{L}\sum_{l=-\infty}^{\infty} g(k_l)f(|k_l|) = \int_0^\infty \mathrm{d}u[g(u)+g(-u)]f(u) + \mathrm{i}\int_0^\infty \mathrm{d}u[f(\mathrm{i}u) - f(-\mathrm{i}u)]\sum_{\chi=\pm1}\frac{g(\mathrm{i}\chi u)}{\mathrm{e}^{Lu+2\pi\mathrm{i}\chi\tilde{\beta}}-1}, \tag{3.4}$$

choosing $g(u) = 1$ and

$$f(u) = \mathrm{Re}\left[K_{1/2+\mathrm{i}m_n\alpha}\left(\mathrm{i}\eta\sqrt{u^2+p^2}\right)K_{1/2+\mathrm{i}m_n\alpha}\left(-\mathrm{i}\eta\sqrt{u^2+p^2}\right)\right]. \tag{3.5}$$

The first term in the rhs of (3.4) is responsible for the azimuthal current in the cosmic string background without compactification, that will be denoted below by $\langle j^2\rangle_s$. The second one corresponds to the contribution due to the compactification of the string along the z-axis, denoted by $\langle j^2\rangle_c$. Therefore, the application of the summation formula (3.4) allows us to decompose the azimuthal current as

$$\langle j^2\rangle_{\rm reg} = \langle j^2\rangle_{s,\rm reg} + \langle j^2\rangle_{c,\rm reg}. \tag{3.6}$$

As we shall see, the compactified part goes to zero in the limit $L \to \infty$.

We start the evaluation with the part corresponding to the geometry of a straight cosmic string, $\langle j^2\rangle_{s,\rm reg}$. Using the first term in the Abel–Plana formula, for this part we get

$$\langle j^2\rangle_{s,\rm reg} = -\frac{2eq\eta^5}{\pi^3\alpha^4 r}\int_0^\infty \mathrm{d}p\, p^2\int_0^\infty \mathrm{d}k \sum_j \epsilon_j J_{\beta_1}(pr)J_{\beta_2}(pr) \times \sum_{n=0,1} c_n\, \mathrm{Re}\left[K_{1/2+\mathrm{i}m_n\alpha}(\mathrm{i}\gamma\eta)K_{1/2+\mathrm{i}m_n\alpha}(-\mathrm{i}\gamma\eta)\right], \tag{3.7}$$

where $\gamma = \sqrt{k^2+p^2}$. Replacing the product of the MacDonald functions by the integral representation [60]

$$K_\nu(\mathrm{i}x)K_\nu(-\mathrm{i}x) = \int_0^\infty \mathrm{d}u u^{-1}\int_0^\infty \mathrm{d}y \cosh(2\nu y)\exp\left[-2ux^2\sinh^2 y - 1/(2u)\right], \tag{3.8}$$

the integral over k is evaluated directly. Performing the integral over p with the help of the formula from [61] we obtain

$$\begin{aligned}\langle j^2\rangle_{s,\mathrm{reg}} = &-\frac{\sqrt{2}eq\eta^5}{r^5\pi^{5/2}\alpha^4}\int_0^\infty \mathrm{d}y\ \cosh y\sum_{n=0,1}c_n\cos(2m_n\alpha y)\\ &\times\int_0^\infty \mathrm{d}z\ z^{3/2}\mathrm{e}^{-z\left[1+2(\eta/r)^2\sinh^2 y\right]}\mathcal{J}(q, a_0, z),\end{aligned} \tag{3.9}$$

where we have introduced the notation

$$\mathcal{J}(q, a_0, z) = \sum_j\left[I_{\beta_1}(z) - I_{\beta_2}(z)\right]. \tag{3.10}$$

Now, in the expression (3.9) the regularization can be safely removed by taking the limit $m_1 \to \infty$. The contribution of the regulator field goes to zero exponentially and, after the integration over y, for the renormalizaed VEV one finds

$$\langle j^2\rangle_s = -\frac{eq\alpha^{-4}}{\sqrt{2}\pi^{5/2}}\int_0^\infty \mathrm{d}x\ x^{3/2}\mathrm{e}^{x\left(1-r^2/\eta^2\right)}\mathcal{J}\left(q, a_0, xr^2/\eta^2\right)\mathrm{Re}\ \left[K_{1/2+\mathrm{i}m\alpha}(x)\right]. \tag{3.11}$$

The current density given by this formula is a periodic odd function of the flux along the string axis with the period equal to the flux quantum. The azimuthal current density, given by (3.11), depends on the radial and time coordinates through the ratio r/η. This property is a consequence of the maximal symmetry of dS spacetime and the Bunch–Davies vacuum. By taking into account that the combination $r_p = \alpha r/\eta$ is the proper distance from the string, we see that r/η is the proper distance measured in units of the dS curvature scale α.

For the further transformation of the expression in the rhs of (3.11), we follow the procedure used in [36] for the summation over j in (3.10). Introducing the notation

$$\mathcal{I}(q, a_0, z) = \sum_j I_{\beta_1}(z), \tag{3.12}$$

we can see that $\sum_j I_{\beta_2}(z) = \mathcal{I}(q, -a_0, z)$. For the series (3.12) one has the integral representation

$$\begin{aligned}\mathcal{I}(q, a_0, z) = &\frac{\mathrm{e}^z}{q} - \frac{1}{\pi}\int_0^\infty \mathrm{d}x\frac{\mathrm{e}^{-z\cosh x}f(q, a_0, x)}{\cosh(qx) - \cos(q\pi)}\\ &+\frac{2}{q}\sum_{k=1}^{p}(-1)^k\cos\left[2\pi k\left(a_0 - 1/(2q)\right)\right]\mathrm{e}^{z\cos(2\pi k/q)},\end{aligned} \tag{3.13}$$

in which p is an integer defined by $2p < q < 2p + 2$ and

$$f(q, a_0, x) = \sum_{\chi=-,+}\chi\cos\left[q\pi\left(1/2 - \chi a_0\right)\right]\cosh\left[\left(qa_0 + \chi q/2 - 1/2\right)x\right]. \tag{3.14}$$

In the specific case $q = 2p$, the term $(-1)^{q/2+1}\sin(q\pi a_0)/(q\mathrm{e}^z)$ should be added to the rhs of (3.13). By taking into account (3.13), for the function (3.10) one gets

$$\mathcal{J}(q, a_0, z) = \frac{4}{\pi}\int_0^\infty \mathrm{d}x \frac{\mathrm{e}^{-z\cosh(2x)} g(q, a_0, 2x)\cosh x}{\cosh(2qx) - \cos(q\pi)} + \frac{4}{q}\sum_{k=1}^{p}(-1)^k \sin(\pi k/q)\sin(2\pi k a_0)\mathrm{e}^{z\cos(2\pi k/q)}, \tag{3.15}$$

where

$$g(q, a_0, x) = \sum_{\chi=-,+} \chi\cos\left[q\pi(1/2 + \chi a_0)\right]\cosh\left[(1/2 - \chi a_0)qx\right]. \tag{3.16}$$

In the case $q = 2p$, the term $2(-1)^{q/2+1}\sin(q\pi a_0)/(q\mathrm{e}^z)$ must be added to the rhs of (3.15). As is seen, in the absence of a magnetic flux along the string, $a_0 = 0$, one has $\mathcal{J}(q, 0, z) = 0$ and the azimuthal current density $\langle j^2\rangle_s$ vanishes.

The azimuthal current density in dS spacetime induced by the magnetic flux is obtained as a special case with $q = 1$. In this case, the expression for the azimuthal current density is simplified to

$$\langle j^2\rangle_s = 2^{3/2}\frac{e\sin(\pi a_0)}{\pi^{7/2}\alpha^4}\int_0^\infty \mathrm{d}u\ \cosh(2a_0 u) \times \int_0^\infty \mathrm{d}x\ x^{3/2}\mathrm{e}^{x\left[1-2(r/\eta)^2\cosh^2 u\right]}\ \mathrm{Re}\left[K_{1/2+\mathrm{i}m\alpha}(x)\right]. \tag{3.17}$$

An equivalent expression for the azimuthal current in general case of q is obtained substituting (3.15) into (3.9), after the removing the regulator part and integrating over z:

$$\langle j^2\rangle_s = -\frac{3e}{4\pi^2(\alpha r/\eta)^4}\int_0^\infty \mathrm{d}u\ \cos(2m\alpha\ \mathrm{arcsinh}\ (ur/\eta)) \times\left[\frac{q}{\pi}\int_0^\infty \mathrm{d}x\frac{g(q, a_0, 2x)\cosh x}{\cosh(2qx) - \cos(q\pi)}\left(\cosh^2 x + u^2\right)^{-5/2} + \sum_{k=1}^{p}\frac{(-1)^k\sin(\pi k/q)\sin(2\pi k a_0)}{\left[\sin^2(\pi k/q) + u^2\right]^{5/2}}\right]. \tag{3.18}$$

In the absence of the conical defect one has $q = 1$ and from (3.18) we find a simpler formula

$$\langle j^2\rangle_s = \frac{3e\sin(\pi a_0)}{4\pi^3\alpha^4}\int_0^\infty \mathrm{d}u\ \cosh(2a_0 u)\int_0^\infty \mathrm{d}x\ \frac{\cos(2m\alpha\ \mathrm{arcsinh}\ x)}{\left[x^2 + (r/\eta)^2\cosh^2 u\right]^{5/2}}. \tag{3.19}$$

For a massless field, integrating over u, from the general formula (3.18) we can see that $\langle j^2\rangle_s$ is conformally related to the corresponding induced current in pure cosmic string spacetime [44], with the conformal factor $(\eta/\alpha)^4$.

In the region near the string, $r/\eta \ll 1$, the dominant contribution to (3.11) comes from large x and we can use the asymptotic expression of the MacDonald function for large arguments. The leading term is independent of the mass and diverges on the string with the inverse fourth power of the proper distance. At large distances from the string, $r/\eta \gg 1$, in (3.11) we use the asymptotic form of the MacDonald function for small values of the argument. The leading term in the asymptotic expansion of the azimuthal current behaves as

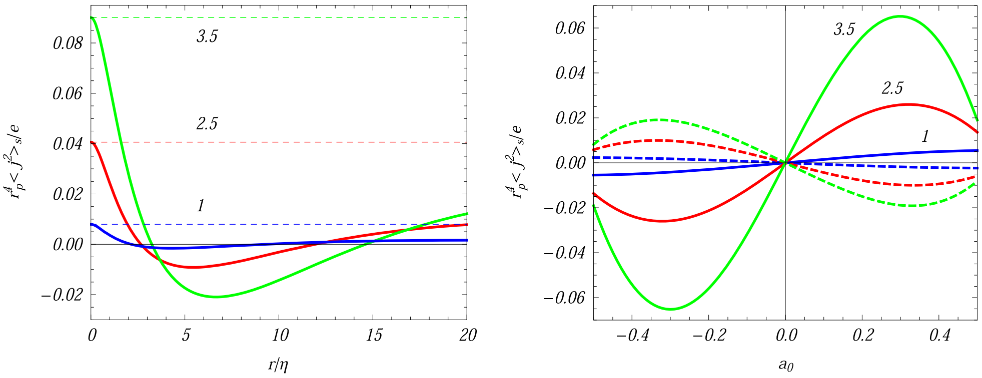

Figure 1. The azimuthal current in the geometry of a straight cosmic string, multiplied by r_p^4, as a function of the distance from the string (left panel) and as a function of the parameter a_0 (right panel) for separate values of q (numbers near the curves). On the left panel the graphs are plotted for $a_0 = 1/4$. The full curves correspond to a massive field with $m\alpha = 1$ and the dashed curves are for a massless field. On the right panel the graphs are plotted for fixed values of $r/\eta = 1$ (full curves) and $r/\eta = 5$ (dashed curves).

$$\langle j^2 \rangle_s \propto (\alpha r/\eta)^{-4} \cos\,[2m\alpha\, \ln(r/\eta) + \theta], \tag{3.20}$$

with a phase θ depending on the parameters $m\alpha$, q and a_0. As a result, at large distances the azimuthal current in the geometry of a straight cosmic string damps oscillatory with the amplitude decaying as the inverse fourth power of the proper distance from the string. The oscillation frequency increases with increasing mass. In the region under consideration the influence of the gravitational field on the current density is essential. The behavior of the current density in dS spacetime, described by (3.20), is crucially different from that for the string in Minkowski bulk. In the latter case, at large distances from the string the current density for a massive field is suppressed exponentially, by the factor $\mathrm{e}^{-2mr\,\sin(\pi/q)}$ for $q \geqslant 2$ and by the factor e^{-2mr} for $q < 2$.

In the left panel of figure 1 we have plotted the quantity $r_p^4 \langle j^2 \rangle_s / e$, with $r_p = \alpha r/\eta$ being the proper distance from the string, as a function of the ratio r/η (proper distance measured in units of α) for $a_0 = 1/4$ and for separate values of the parameter q (the numbers near the curves). The full curves correspond to a massive field with $m\alpha = 1$ and the dashed lines are for a massless field. In the latter case the combination $r_p^4 \langle j^2 \rangle_s$ does not depend on r/η. From the asymptotic analysis given above it follows that $r_p^4 \langle j^2 \rangle_s |_{r=0} = r_p^4 \langle j^2 \rangle_s |_{m=0}$ which is also seen from the graphs. For large values of r/η and for a massive field we see the characteristic oscillations described by (3.20). In the right panel of figure 1 the quantity $r_p^4 \langle j^2 \rangle_s / e$ is displayed as a function of the parameter a_0 for fixed values $r/\eta = 1$ (full curves) and $r/\eta = 5$ (dashed curves). Again, the numbers near the full curves correspond to the values of q and we have taken $m\alpha = 1$. For the dashed curves the same values of q are used and $|\langle j^2 \rangle_s|$ increases with increasing q.

The azimuthal current density (3.18) generates a magnetic field around a straight cosmic string directed parallel to the axis of the string. A qualitatively new feature for a string in dS bulk is the time-dependence of this magnetic field. The latter means that, in addition to the magnetic field, the vacuum current generates also an electric field along the azimuthal direction. At large distances from the string, the electromagnetic fields decay as power-law for both massless and massive fermionic fields, giving rise to long-range effects. For a string in flat spacetime, the electric field is absent and in the massive case the magnetic field is exponentially small at distances larger than the Compton wavelength of the fermionic particle.

Now, we turn to the part in the azimuthal current density induced by the compactification of the string along its axis. In this part, due the presence of the exponential factor in the denominator of the integrand, the contribution of the regulator field may be removed from the beginning by taking the limit $m_1 \to \infty$. Using the second term in the rhs of the Abel–Plana formula and the expansion $\left(e^u - 1\right)^{-1} = \sum_{l=1}^{\infty} e^{-lu}$, from (3.3) for the topological part we obtain

$$\langle j^2 \rangle_c = -\frac{2eq\eta^5}{\pi^2\alpha^4 r}\sum_{l=1}^{\infty}\cos\left(2\pi\tilde{\beta}l\right)\int_0^\infty \mathrm{d}p\, p^2 \sum_j \epsilon_j J_{\beta_1}(pr)J_{\beta_2}(pr)\int_0^\infty \mathrm{d}\lambda\,\lambda \\ \times \frac{e^{-lL\sqrt{\lambda^2+p^2}}}{\sqrt{\lambda^2+p^2}}\,\mathrm{Re}\,\left\{K_{1/2+\mathrm{i}m\alpha}(\eta\lambda)\left[I_{-1/2-\mathrm{i}m\alpha}(\eta\lambda)+I_{1/2+\mathrm{i}m\alpha}(\eta\lambda)\right]\right\}. \tag{3.21}$$

For the further transformation we employ the integral representation (see also [49])

$$\frac{e^{-lL\sqrt{\lambda^2+p^2}}}{\sqrt{\lambda^2+p^2}} = \frac{2}{\sqrt{\pi}}\int_0^\infty \mathrm{d}s\; e^{-\left(\lambda^2+p^2\right)s^2-l^2L^2/4s^2}. \tag{3.22}$$

Substituting into (3.21), after the integrations over p and λ, for the topological contribution in the azimuthal current density we get

$$\langle j^2 \rangle_c = -\frac{\sqrt{2}\,eq}{\pi^{5/2}\alpha^4}\sum_{l=1}^{\infty}\cos\left(2\pi\tilde{\beta}l\right)\int_0^\infty \mathrm{d}x\; x^{3/2}e^{x\left[1-r^2/\eta^2-l^2L^2/\left(2\eta^2\right)\right]} \\ \times \mathcal{J}\left(q, a_0, xr^2/\eta^2\right)\mathrm{Re}\,\left[K_{1/2+\mathrm{i}m\alpha}(x)\right], \tag{3.23}$$

with the function $\mathcal{J}(q, a_0, z)$ given by (3.15). As is seen from this expression, the topological part in the azimuthal current density is a periodic odd function of the magnetic flux along the string and a periodic even function of the flux enclosed by the compactified dimension. In both cases, the period is equal to the flux quantum.

By taking into account (3.11), the total azimuthal current is written in the form

$$\langle j^2 \rangle = -\frac{\sqrt{2}\,eq}{\pi^{5/2}\alpha^4}{\sum_{l=0}^{\infty}}'\cos\left(2\pi\tilde{\beta}l\right)\int_0^\infty \mathrm{d}x\; x^{3/2}e^{x\left[1-r^2/\eta^2-l^2L^2/\left(2\eta^2\right)\right]} \\ \times \mathcal{J}\left(q, a_0, xr^2/\eta^2\right)\mathrm{Re}\,\left[K_{1/2+\mathrm{i}m\alpha}(x)\right], \tag{3.24}$$

where the prime on the sign of the summation means that the term $l = 0$ should be taken with the coefficient 1/2. The latter corresponds to $\langle j^2 \rangle_s$. The azimuthal current density depends on L, r, and η through the ratios L/η and r/η which are the proper length of the compact dimension and proper distance from the string axis measured in units of α. Again, this feature is a consequence of the maximal symmetry of dS spacetime. In the absence of the planar angle deficit one has $q = 1$ and the general formula (3.24) is reduced to the expression

$$\langle j^2 \rangle = \frac{4\sqrt{2}e}{\pi^{7/2}\alpha^4} \sin(\pi a_0) \sum_{l=0}^{\infty}{}' \cos\left(2\pi\tilde{\beta}l\right) \int_0^\infty du \ \cosh\left(2a_0 u\right) \times \int_0^\infty dx \ x^{3/2} e^{x\left[1-2(r/\eta)^2\cosh^2 u - l^2L^2/\left(2\eta^2\right)\right]} \ \mathrm{Re}\ \left[K_{1/2+im\alpha}(x)\right]. \tag{3.25}$$

The latter presents the azimuthal current induced by an infinitely thin flux tube along the compactified z-axis.

For a massless field, by taking into account that $K_{1/2}(x) = e^{-x}\sqrt{\pi/2x}$ and using (3.15), the integration over x in (3.24) is done explicitly and one finds

$$\langle j^2 \rangle = -\frac{16\pi^{-2}e}{(\alpha/\eta)^4} \sum_{l=0}^{\infty}{}' \cos\left(2\pi\tilde{\beta}l\right) \left[\sum_{k=1}^{p} \frac{(-1)^k \sin(\pi k/q) \sin\left(2\pi k a_0\right)}{\left[4r^2 \sin^2(\pi k/q) + l^2L^2\right]^2} + \frac{q}{\pi}\int_0^\infty dx \frac{g\left(q, a_0, 2x\right)\cosh x}{\cosh(2qx) - \cos(q\pi)} \left(4r^2\cosh^2 x + l^2L^2\right)^{-2}\right]. \tag{3.26}$$

In this case the azimuthal current is equal to $(\eta/\alpha)^4$ times the one for the flat space in the presence of the compactified cosmic string [44].

Let us discuss the behavior of the topological part in the azimuthal current in the asymptotic regions of the parameters. Near the cosmic string, to the leading order, for the topological contribution we get

$$\langle j^2 \rangle_c = \frac{\mathrm{sgn}\ (a_0)\sqrt{2}eq\left(r/\sqrt{2}\eta\right)^{q\left(1-2|a_0|\right)-1}}{\pi^{5/2}\alpha^4\Gamma\left(q\left(1/2-|a_0|\right)+1/2\right)} \sum_{l=1}^{\infty} \cos\left(2\pi\tilde{\beta}l\right) \times \int_0^\infty dx\ x^{q\left(1/2-|a_0|\right)+1} e^{x\left[1-l^2L^2/\left(2\eta^2\right)\right]}\ \mathrm{Re}\ \left[K_{1/2+im\alpha}(x)\right]. \tag{3.27}$$

From here it follows that the topological part in the azimuthal current vanishes on the string for $|a_0| < (1-1/q)/2$, is finite for $|a_0| = (1-1/q)/2$ and diverges for $|a_0| > (1-1/q)/2$. Note that in the latter case the divergent contribution comes from the irregular mode. For $a_0 \neq 0$, in the region near the string the total current is dominated by the part $\langle j^2 \rangle_s$.

For large values r, $r/\eta \gg 1$, the dominant contribution to the integral in (3.24) comes from the values of x in the region $x \lesssim \eta^2/r^2$ and we replace the MacDonald function by its asymptotic form for small values of the argument. At large distances, the behavior of the azimuthal current density depends crucially on whether the parameter $\tilde{\beta}$ is zero or not. For $0 < \tilde{\beta} < 1$ and $q \geqslant 2$, to the leading order one finds

$$\langle j^2 \rangle \approx \frac{2e\sigma_\beta\eta^2/L^2}{\alpha^4(r/\eta)^2} \frac{\sin\left(2\pi a_0\right)}{\sin(\pi/q)} \frac{e^{-4\pi\sigma_\beta r \sin(\pi/q)/L}}{\sqrt{\cosh(\pi m\alpha)}} \times \cos\left[m\alpha \ln\left(\frac{2rL\sin(\pi/q)}{\pi\sigma_\beta\eta^2}\right) + \beta_0(m\alpha)\right], \tag{3.28}$$

where β_0 is the phase of the function $\Gamma(1/2 + im\,\alpha)$ and $\sigma_\beta = \min(\tilde{\beta}, 1-\tilde{\beta})$. Hence, for $\tilde{\beta} \neq 0$, at large distances the topological part in the current density is exponentially small. In the case $q < 2$ the suppression is stronger, by the factor $e^{-4\pi\sigma_\beta r/L}$.

At large distances, $r/\eta \gg 1$, and for $\tilde{\beta} = 0$, in (3.24) the dominant contribution to the series comes from large l and the MacDonald function is replaced by its asymptotic form for small values of the argument. After the integration over x we get

$$\langle j^2 \rangle \approx -\frac{eq\eta B_q(a_0, m\alpha)}{\pi^3\alpha^4 L(r/\eta)^3} \cos\Big[2m\alpha \ln(2r/\eta) + \beta_q(a_0, m\alpha)\Big], \tag{3.29}$$

where the functions $B_q(a_0, m\alpha)$ and $\beta_q(a_0, m\alpha)$ are defined by the relation

$$\begin{aligned} B_q(a_0, m\alpha)\mathrm{e}^{\mathrm{i}\beta_q(a_0,m\alpha)} &= \Gamma(1/2 + \mathrm{i}m\alpha)\Gamma(3/2 - \mathrm{i}m\alpha) \\ &\times \Bigg\{ \int_0^\infty \mathrm{d}x \frac{g(q, a_0, 2x)(\cosh x)^{2\mathrm{i}m\alpha-2}}{\cosh(2qx) - \cos(q\pi)} \\ &+ \frac{\pi}{q} \sum_{k=1}^{p} \frac{(-1)^k \sin(2\pi k a_0)}{[\sin(\pi k/q)]^{2-2\mathrm{i}m\alpha}} \Bigg\}. \end{aligned} \tag{3.30}$$

In this case the decay is of power-law for both massive and massless fields. This is in clear contrast with the case of the cosmic string in flat spacetime where the decay of the topological part of the current density for massive fields is exponential.

For large values L/η and for fixed r/η, i.e. $L \gg r$, the dominant contribution in the integral of (3.23) comes from the region near the lower limit of the integration. Expanding in the integrand over x , to the leading order we find

$$\begin{aligned} \langle j^2 \rangle_c \approx &\frac{4eqB_2(m\alpha)}{\pi^{5/2}(\alpha L/\eta)^4} \frac{\mathrm{sgn}\,(a_0)(r/L)^{q(1-2|a_0|)-1}}{\Gamma\big(q\big(1/2 - |a_0|\big) + 1/2\big)} \\ &\times \sum_{l=1}^{\infty} \frac{\cos\big(2\pi\tilde{\beta}l\big)}{l^{q(1-2|a_0|)+3}} \cos\Big[2m\alpha \ln(lL/\eta) + \beta_2(m\alpha)\Big], \end{aligned} \tag{3.31}$$

where the functions $B_2(m\alpha)$ and $\beta_2(m\alpha)$ are defined by the relation

$$B_2(m\alpha)\mathrm{e}^{\mathrm{i}\beta_2(m\alpha)} = \Gamma(1/2 + \mathrm{i}m\alpha)\Gamma\Big(q\big(1/2 - |a_0|\big) + 3/2 - \mathrm{i}m\alpha\Big), \tag{3.32}$$

with $B_2(m\alpha)$ being the modulus of the expression on the right. Note that in the geometry of a cosmic string on Minkowski bulk the VEV of the azimuthal current density for large values of L is suppressed exponentially, by the factor e^{-mL}.

For small values of L and for $0 < \tilde{\beta} < 1$, it can be seen that the current density $\langle j^2 \rangle$ is suppressed by the factor $\mathrm{e}^{-4(\pi\sigma_\beta r/L)\sin(\pi/q)}$ for $q \geqslant 2$ and by the factor $\mathrm{e}^{-4\pi\sigma_\beta r/L}$ for $q < 2$. For $\tilde{\beta} = 0$, to the leading order we get

$$\langle j^2 \rangle \approx -\frac{eq\eta}{\pi^2\alpha^4 L} \int_0^\infty \mathrm{d}x\, x\mathrm{e}^{x(1-r^2/\eta^2)} \mathcal{J}\big(q, a_0, xr^2/\eta^2\big) \mathrm{Re}\,\big[K_{1/2+\mathrm{i}m\alpha}(x)\big]. \tag{3.33}$$

Note that in this limit the topological part dominates in the azimuthal current density.

In figure 2, we have presented the dependence of the azimuthal current density, multiplied by r_p^4, on the proper distance from the z-axis measured in units of α, r/η, for separate values of the ratio L/η (numbers near the curves). The dashed curves present the corresponding quantity in the geometry of a uncompactified magnetic flux along the z-axis, $r_p^4 \langle j^2 \rangle_s / e$. The specific values of the parameters are chosen as follows: $q = 1$, $m\alpha = 1$, $a_0 = 1/4$. For the left panel $\tilde{\beta} = 0$ and for the right one $\tilde{\beta} = 1/2$. The features of the asymptotic analysis described above are clearly seen in the numerical examples presented. In

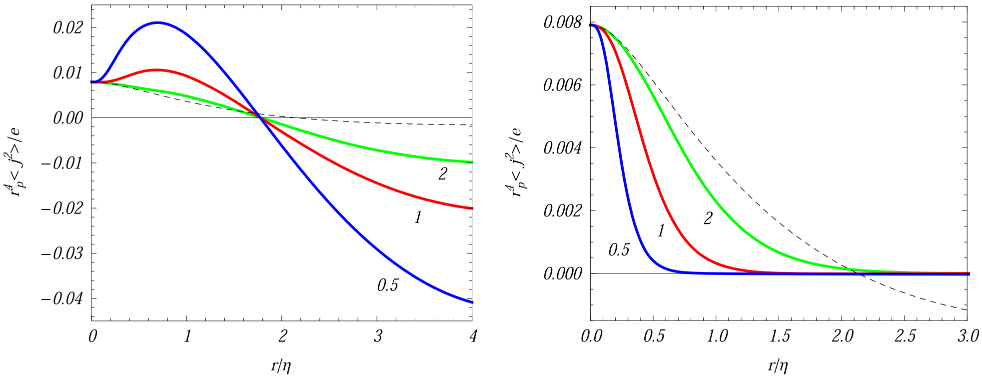

Figure 2. The azimuthal current density, multiplied by r_p^4, as a function of the distance from the z-axis for $\tilde{\beta} = 0$ (left panel) and $\tilde{\beta} = 1/2$ (right panel). The values of the other parameters are chosen as follows: $q = 1$, $m\alpha = 1$, $a_0 = 1/4$. The dashed curves correspond to the azimuthal current in the geometry of an uncompactified magnetic flux. The numbers near the curves correspond to the values of L/η.

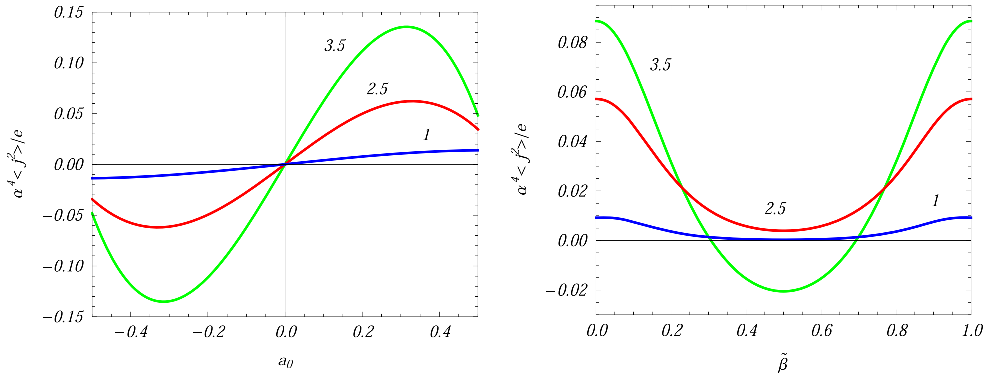

Figure 3. The azimuthal current density as a function of a_0 (left panel) and $\tilde{\beta}$ (right panel) for different values of the parameter q (numbers near the curves). The graphs are plotted for $m\alpha = 1$ and $L/\eta = r/\eta = 1$. For the left panel $\tilde{\beta} = 0$ and for the right one $a_0 = 1/4$.

particular, for $\tilde{\beta} = 0$ the topological part dominates for large r/η, whereas for $\tilde{\beta} = 1/2$ the current density is suppressed.

In figure 3, the azimuthal current density is displayed as a function of a_0 (left panel) and $\tilde{\beta}$ (right panel) for different values of the parameter q (numbers near the curves). The graphs are plotted for the values of the parameters $m\alpha = 1$ and $L/\eta = r/\eta = 1$. For the left panel we have taken $\tilde{\beta} = 0$ and for the right one $a_0 = 1/4$.

4. Axial current

In this section we investigate the axial component of the current density. In order to do that we substitute the mode functions (2.13) and the relevant gamma matrices into (2.12) with $\mu = 3$. After some intermediate steps the regularized current density is presented in the form

$$\langle j^3\rangle_{\rm reg} = -\frac{e q\eta^5}{\pi^2 L\alpha^4}\int_0^\infty {\rm d}p\, p\, \sum_j \Big[J^2_{\beta_1}(pr) + J^2_{\beta_2}(pr)\Big] \times \sum_{n=0,1} c_n \sum_{l=-\infty}^{+\infty} k_l\ {\rm Re}\,\Big[K_{1/2+{\rm i}m_n\alpha}({\rm i}\gamma\eta)K_{1/2+{\rm i}m_n\alpha}(-{\rm i}\gamma\eta)\Big], \tag{4.1}$$

where we have replaced the Hankel functions by the MacDonald ones. The physical origin of the axial current is the nontrivial phase in the periodicity condition (2.8) (or a constant gauge field in the gauge transformed representation). The expression in the right-hand side of (4.1) is a periodic function of $\tilde{\beta}$ with the period 1 and, without loss of generality, we can assume that $0 \leqslant \tilde{\beta} < 1$. For $\tilde{\beta} = 0$ the momenta k_l of the modes with $l = |l|$ and $l = -|l|$ have opposite signs and their contributions in (4.1) cancel each other. As a result, in this case the axial current vanishes. The allowance of the nontrivial phase $\tilde{\beta} \neq 0$ breaks the symmetry between the modes with opposite signs of the quantum number l. Now the corresponding momenta k_l does not compensate each other and the net vacuum current appears. In this sense, the physical reason for the appearance of the nonzero VEV for the current density along the compact dimension is similar to that for persistent currents in mesoscopic metallic rings threaded by a magnetic flux [62]. This type of currents have been recently observed experimentally [63].

To perform the summation over l in the expression (4.1), we use again the Abel–Plana formula (3.4) as we did for $\langle j^2\rangle$, taking $g(u) = u$ and the expression given in (3.5) for $f(u)$. The function $g(u)$ is an odd function which means that the first term on the rhs of the Abel–Plana formula vanishes. Therefore, the axial current in the geometry of a straight cosmic string vanishes.

The contribution to the axial current induced by the compactification is given by the second term in the rhs of the summation formula (3.4). In this part the contribution coming from the regulator field can be removed from the beginning. By using the expansion $\left({\rm e}^u - 1\right)^{-1} = \sum_{l=1}^{\infty} {\rm e}^{-lu}$, the topological part is written in the form

$$\langle j^3\rangle = -\frac{e q\eta^5}{\pi^2\alpha^4}\sum_{l=1}^{\infty}\sin\!\left(2\pi\tilde{\beta}l\right)\int_0^\infty {\rm d}p\, p\, \sum_j \Big[J^2_{\beta_1}(pr) + J^2_{\beta_2}(pr)\Big]\int_0^\infty {\rm d}\lambda\, \lambda \times {\rm e}^{-lL\sqrt{\lambda^2+p^2}}\ {\rm Re}\,\Big\{K_{1/2+{\rm i}m\alpha}(\lambda\eta)\Big[I_{-1/2-{\rm i}m\alpha}(\lambda\eta) + I_{1/2+{\rm i}m\alpha}(\lambda\eta)\Big]\Big\}. \tag{4.2}$$

As the next step, we use the integral representation

$$ {\rm e}^{-lL\sqrt{\lambda^2+p^2}} = \frac{lL}{\sqrt{\pi}}\int_0^\infty {\rm d}s\, s^{-2}{\rm e}^{-\left(\lambda^2+p^2\right)s^2 - l^2L^2/4s^2}. \tag{4.3}$$

Substituting into (4.2) and changing the order of integrations, the integrals over λ and p are evaluated and the final result is written as

$$\langle j^3\rangle = -\frac{\sqrt{2}\,eL}{\pi^{5/2}\alpha^4}\sum_{l=1}^{\infty} l\,\sin\!\left(2\pi\tilde{\beta}l\right)\int_0^\infty {\rm d}x\, x^{3/2}\,{\rm e}^{x\left(1-l^2L^2/2\eta^2\right)} \times {\rm Re}\,\Big[K_{1/2+{\rm i}m\alpha}(x)\Big]\mathcal{F}\!\left(q, a_0, xr^2/\eta^2\right), \tag{4.4}$$

with the function

$$\mathcal{F}(q, a_0, z) = \frac{q}{2}{\rm e}^{-z}\big[\mathcal{I}(q, a_0, z) + \mathcal{I}(q, -a_0, z)\big]. \tag{4.5}$$

For the latter, by using the representation (3.13) of the function $\mathcal{I}(q, a_0, z)$, one gets

$$\mathcal{F}(q, a_0, z) = 1 + 2\sum_{k=1}^{p}(-1)^k \cos\left(2\pi k a_0\right)\frac{\cos(\pi k/q)}{\mathrm{e}^{2z\sin^2(\pi k/q)}} + \frac{2q}{\pi}\int_0^\infty \mathrm{d}x \frac{\mathrm{e}^{-2z\cosh^2 x} h\left(q, a_0, 2x\right)\sinh x}{\cosh(2qx) - \cos(q\pi)}, \tag{4.6}$$

with the notation

$$h(q, a_0, x) = \sum_{\chi=\pm 1}\cos\left[\left(1/2 + \chi a_0\right)q\pi\right]\sinh\left[\left(1/2 - \chi a_0\right)qx\right]. \tag{4.7}$$

The axial current is an odd function of the parameter $\tilde{\beta}$ and an even function of a_0.

The part in the current density coming from the first term in the right-hand side of (4.6),

$$\langle j^3\rangle_0 = -\frac{\sqrt{2}\,eL}{\pi^{5/2}\alpha^4}\sum_{l=1}^{\infty} l\sin\left(2\pi\tilde{\beta}l\right)\int_0^\infty \mathrm{d}x\, x^{3/2}\,\mathrm{e}^{x\left(1 - l^2L^2/2\eta^2\right)}\,\mathrm{Re}\left[K_{1/2+\mathrm{i}m\alpha}(x)\right], \tag{4.8}$$

does not depend on the planar angle deficit and magnetic flux along the string axis. It is a purely topological contribution and coincides with the current density in dS spacetime with spatial topology $R^2 \times S^1$ in the absence of the string and of the magnetic flux along the z-axis. The fermionic current in $(D+1)$-dimensional dS spacetime with topology $R^p \times (S^1)^{D-p}$ has been investigated in [47]. By using the integral representation for the function $K_\nu(x)$, it can be seen that (4.8) coincides with the result from [47] in the special case $D = 3$, $p = 2$. The part in the axial current with the second and third terms on the right of (4.6) are induced by the presence of the string and of the magnetic flux along its axis.

In the absence of the cosmic string one has $q = 1$ and from (4.4) we get

$$\langle j^3\rangle = \langle j^3\rangle_0 - \frac{2^{3/2}eL\sin\left(a_0\pi\right)}{\pi^{7/2}\alpha^4}\sum_{l=1}^{\infty} l\sin\left(2\pi\tilde{\beta}l\right)\int_0^\infty \mathrm{d}u\ \sinh\left(2a_0 u\right) \times \tanh(u)\int_0^\infty \mathrm{d}x\, x^{3/2}\,\mathrm{e}^{x\left[1 - 2(r/\eta)^2\cosh^2 u - l^2L^2/2\eta^2\right]}\,\mathrm{Re}\left[K_{1/2+\mathrm{i}m\alpha}(x)\right]. \tag{4.9}$$

The second term on the rhs of this formula is induced by infinitely thin magnetic flux running along the compactified z-axis.

For a massless Dirac field the MacDonald function in the general formula (4.4) is expressed in terms of the exponential function. Substituting (4.6) into (4.4), after the integration over z we find

$$\langle j^3\rangle = -\frac{4e(\eta/\alpha)^4}{\pi^2L^3}\sum_{l=1}^{\infty} l\sin\left(2\pi\tilde{\beta}l\right)\left[\frac{1}{l^4} + 2\sum_{k=1}^{p}\frac{(-1)^k\cos\left(2\pi k a_0\right)\cos(\pi k/q)}{\left[l^2 + 4(r/L)^2\sin^2(\pi k/q)\right]^2} + \frac{2q}{\pi}\int_0^\infty \mathrm{d}x\frac{h\left(q, a_0, 2x\right)\sinh x}{\cosh(2qx) - \cos(q\pi)}\left[l^2 + 4(r/L)^2\cosh^2 x\right]^{-2}\right]. \tag{4.10}$$

In this case the induced axial current is equal to $(\eta/\alpha)^4$ times the corresponding one in Minkowski spacetime in the presence of the cosmic string (the sign of the axial current obtained in [44] for the string in Minkowski bulk should be corrected to the opposite one).

Now let us consider the general formula (4.4) in various asymptotic regions of the parameters. At large distances from the string, $r/\eta \gg 1$, by taking into account that to the leading order $\mathcal{F}(q, a_0, xr^2/\eta^2) \approx 1$, we see that the current density coincides with the

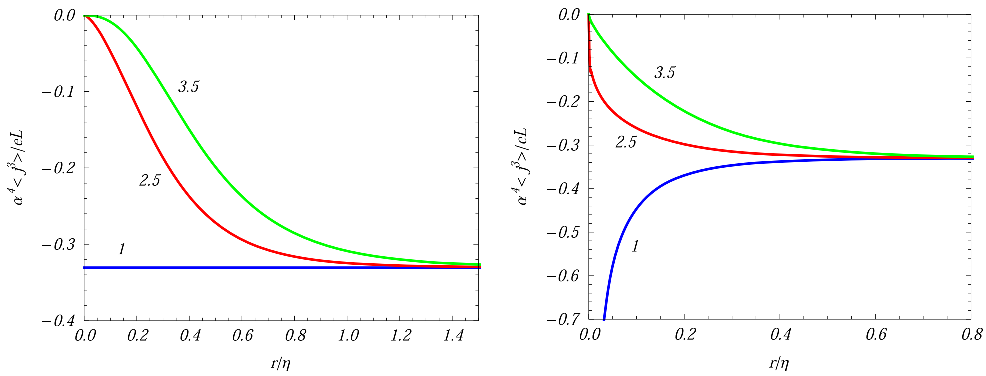

Figure 4. The axial current density as a function of r/η for $a_0 = 0$ (left panel) and $a_0 = 1/4$ (right panel) and for separate values of q (numbers near the curves). The graphs are plotted for $m\alpha = 1$ and $\tilde{\beta} = 1/4$.

corresponding result in dS spacetime when the string and the magnetic flux along the z-axis are absent. In the region near the string, $r/\eta \ll 1$, to the leading order, one gets

$$\langle j^3 \rangle = -\frac{eqL}{\sqrt{2}\pi^{5/2}\alpha^4} \frac{\left(r/\sqrt{2}\eta\right)^{q\left(1-2|a_0|\right)-1}}{\Gamma\left(q\left(1/2 - |a_0|\right) + 1/2\right)} \sum_{l=1}^{\infty} l \sin\left(2\pi\tilde{\beta}l\right) \times \int_0^{\infty} \mathrm{d}x\, x^{q\left(1/2-|a_0|\right)+1} \mathrm{e}^{x\left(1-l^2L^2/2\eta^2\right)} \operatorname{Re}\left[K_{1/2+\mathrm{i}m\alpha}(x)\right]. \tag{4.11}$$

The axial current vanishes on the string for $|a_0| < (1 - 1/q)/2$, is finite for $|a_0| = (1 - 1/q)/2$ and diverges for $|a_0| > (1 - 1/q)/2$. The irregular mode is responsible for the divergence in the latter case.

For small values of L/η, the dominant contribution to the axial current density comes from the part $\langle j^3 \rangle_0$, corresponding to the current density in the geometry without the cosmic string and magnetic flux along the z-axis, and the contribution of the string-induced part is exponentially suppressed. By taking into account that in the limit under consideration the contribution of large x dominates in the integral of (4.8), to the leading order we find

$$\langle j^3 \rangle \approx -\frac{4\pi^{-2}e}{(\alpha/\eta)^4 L^3} \sum_{l=1}^{\infty} \frac{\sin\left(2\pi\tilde{\beta}l\right)}{l^3}. \tag{4.12}$$

For large values L/η with r/η fixed, in the integral of (4.4) the contribution from the region near the lower limit dominates and for the VEV of the axial current we get

$$\langle j^3 \rangle \approx -\frac{2eqLB_2(m\alpha)(r/L)^{q\left(1-2|a_0|\right)-1}}{\pi^{5/2}(\alpha L/\eta)^4 \Gamma\left(q\left(1/2 - |a_0|\right) + 1/2\right)} \times \sum_{l=1}^{\infty} \frac{\sin\left(2\pi\tilde{\beta}l\right)}{l^{q\left(1-2|a_0|\right)+2}} \cos\left[2m\alpha \ln(lL/\eta) + \beta_2(m\alpha)\right], \tag{4.13}$$

where the functions $B_2(m\alpha)$ and $\beta_2(m\alpha)$ are defined by the relation (3.32). As is seen, for the length of the compact dimension larger than the curvature radius of dS spacetime the axial

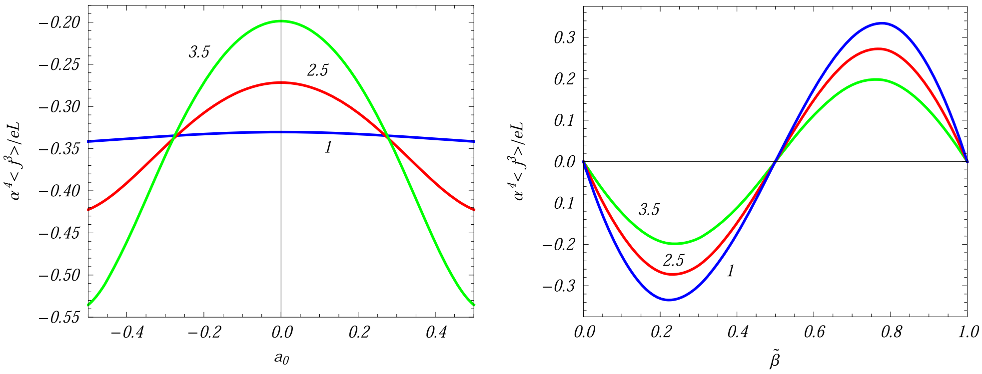

Figure 5. The axial current density versus a_0 (left panel) and $\tilde{\beta}$ (right panel) for different values of the parameter q (the numbers near the curves). For the left panel $\tilde{\beta} = 1/4$ and for the right one $a_0 = 0$. The graphs are plotted for $L/\eta = 1$, $r/\eta = 0.5$, $m\alpha = 1$.

current decays as a power-law. For a string in background of Minkowski spacetime and for a massive fermionic field the axial current density decays as e^{-mL}.

In figure 4 the axial current density is displayed as a function of the ratio r/η for $a_0 = 0$ (left panel) and $a_0 = 1/4$ (right panel). The numbers near the curves are the corresponding values of the parameter q. In both panels the graphs are plotted for $m\alpha = 1$ and $\tilde{\beta} = 1/4$. As it has been explained before, at large distances the axial current density tends to the limiting value corresponding to the geometry without the cosmic string and in the absence of the magnetic flux along the z-axis (the horizontal line in the left panel corresponding to $q = 1$ and $a_0 = 0$).

The figure 5 presents the dependence of the axial current density on the parameters a_0 (left panel) and $\tilde{\beta}$ (right panel) for separate values of q (numbers near the curves). For the left panel we have taken $\tilde{\beta} = 1/4$ and for the right panel $a_0 = 0$. The values of the other parameters are as follows: $L/\eta = 1$, $r/\eta = 0.5$, $m\alpha = 1$. Note that for fixed values of the other parameters the modulus of the axial current, $|\langle j^3 \rangle|$, increases with increasing q for a_0 close to $\pm 1/2$ and decreases for a_0 close to 0.

5. Generalization to higher dimensions

By taking into account possible applications of this analysis to Kaluza–Klein type models, in this section we give the formulas for the current density in $(D + 1)$-dimensional spacetime described by the line element

$$ds^2 = dt^2 - e^{2t/\alpha}\left(dr^2 + r^2 d\phi^2 + \sum_{i=3}^{D}\left(dx^i\right)^2\right), \tag{5.1}$$

considering the compactification along the direction $x^3 = z$. In the irreducible representation, the corresponding Dirac matrices are $N \times N$ matrices with $N = 2[(D + 1)/2]$, where the square brackets mean the integer part. The time-dependent factors in the corresponding mode functions have the form $\eta^{(D+1)/2} H^{(\lambda_\pm)}_{\pm 1/2 - im\alpha}(\gamma\eta)$. The dependence on the coordinates $\mathbf{x}_\parallel = (x^4, \ldots, x^D)$ can be taken in the form $e^{i\mathbf{k}_\parallel \mathbf{x}_\parallel}$ with $\mathbf{k}_\parallel = (k_{\parallel 4}, \ldots, k_{\parallel D})$, $-\infty < k_{\parallel i} < +\infty$, and one has $\gamma = \sqrt{k^2 + k_{\parallel}^2 + p^2}$. The VEV of the current density is given by (2.12), where $\sum_{\sigma}$ now contains also the integration over $\mathbf{k}_{\parallel}$.

The components of the renormalized current density along additional dimensions vanish, $\langle j^{\mu} \rangle = 0$ for $\mu = 4, \ldots, D$. The main steps of the evaluation for the azimuthal and axial currents are essentially the same we have described for the case $D = 3$. By the use of integral representation (3.8), the integration over $\mathbf{k}_{\parallel}$ is reduced to the integral of the form $\int_0^{\infty} \mathrm{d}k_{\parallel} k_{\parallel}{}^{D-4} \mathrm{e}^{-bk_{\parallel}^2}$ which is expressed in terms of the gamma function. For the azimuthal current density one gets the expression

$$\langle j^2 \rangle = -\frac{2eqN\alpha^{-D-1}}{(2\pi)^{D/2+1}} \sideset{}{'}\sum_{l=0}^{\infty} \cos\left(2\pi\tilde{\beta}l\right) \int_0^{\infty} \mathrm{d}x\, x^{D/2} \mathrm{e}^{x\left[1 - r^2/\eta^2 - l^2L^2/\left(2\eta^2\right)\right]} \times \mathcal{J}\left(q, a_0, xr^2/\eta^2\right) \mathrm{Re}\left[K_{1/2+\mathrm{i}m\alpha}(x)\right], \tag{5.2}$$

where the term $l = 0$ corresponds to the azimuthal current density in the geometry of straight cosmic string and the remaining part comes from the compactification. An equivalent expression for the former, that generalizes (3.18), is given by the formula

$$\langle j^2 \rangle_s = -\frac{4DeN\Gamma(D/2)}{(4\pi)^{D/2+1}(\alpha r/\eta)^{D+1}} \int_0^{\infty} \mathrm{d}u \cos(2m\alpha \,\mathrm{arcsinh}\,(ur/\eta)) \times \left[\frac{q}{\pi} \int_0^{\infty} \mathrm{d}x \frac{g\left(q, a_0, 2x\right)\cosh x}{\cosh(2qx) - \cos(q\pi)} \left(\cosh^2 x + u^2\right)^{-D/2-1} + \sum_{k=1}^{p} (-1)^k \frac{\sin(\pi k/q) \sin\left(2\pi k a_0\right)}{\left[\sin^2(\pi k/q) + u^2\right]^{D/2+1}}\right]. \tag{5.3}$$

For a massless field from here we get

$$\langle j^2 \rangle_s = -\frac{2eN\Gamma((D+1)/2)}{(4\pi)^{(D+1)/2}(\alpha r/\eta)^{D+1}} \left[\sum_{k=1}^{p} (-1)^k \frac{\sin\left(2\pi k a_0\right)}{\sin^D(\pi k/q)} + \frac{q}{\pi} \int_0^{\infty} \mathrm{d}x \frac{g\left(q, a_0, 2x\right)\cosh^{-D} x}{\cosh(2qx) - \cos(q\pi)}\right]. \tag{5.4}$$

Though, in deriving (5.2)–(5.4) we have assumed that $D \geqslant 3$, by the analytical continuation they are valid for $D = 2$ as well. In particular, we can see that in the Minkowskian limit and for $D = 2$, formula (5.3) coincides with the expression for the current density derived in [36] (the parameter a_0 here corresponds to the parameter $-\alpha_0$ in [36]).

In a similar way, for the axial current density in an arbitrary number of spatial dimensions we find the expression

$$\langle j^3 \rangle = -\frac{2eNL\alpha^{-D-1}}{(2\pi)^{D/2+1}} \sum_{l=1}^{\infty} l \sin\left(2\pi\tilde{\beta}l\right) \int_0^{\infty} \mathrm{d}x\, x^{D/2} \mathrm{e}^{x\left[1 - l^2L^2/\left(2\eta^2\right)\right]} \times \mathcal{F}\left(q, a_0, xr^2/\eta^2\right) \mathrm{Re}\left[K_{1/2+\mathrm{i}m\alpha}(x)\right]. \tag{5.5}$$

The main features in the behavior of the current density, described above for the case $D = 3$, remain valid for general D. The simplest application for the formulas in this section corresponds to the $D = 4$ Kaluza–Klein model with a single compact dimension. In this model, the induced current along the compact dimension is a source of magnetic field with the components along uncompactified dimensions. Note that we could consider more general problem with additional compact dimension x^i, $i \geqslant 4$. In these models, the application of the summation formula (3.4) allows one to obtain a recurrence formula relating the current density in the model with s compact dimensions to the corresponding quantity for $s - 1$ compact dimensions.

6. Conclusion

In this paper we have investigated the combined effects of the background gravitational field and topology on the VEV of the current density for a massive fermionic field. This quantity is an important local characteristic of the quantum vacuum. In addition to describing the physical structure of a charged quantum field at a given point, the VEV of the current density acts as the source in the semiclassical Maxwell equations and plays an important role in modeling a self-consistent dynamics involving the electromagnetic field. In order to have an exactly solvable problem, we have taken dS spacetime as the background geometry. The latter is among the most popular gravitational backgrounds and plays an important role in cosmology. The topological effects are induced by a cosmic string and by its compactification along the axis. Additionally, we have assumed the presence of a constant gauge field with nonzero axial and azimuthal components. By a gauge transformation, the problem is reduced to the one in the absence of a gauge field with the quasiperiodicity conditions (2.7) and (2.8) on the field operator. In the new representation, the information about the gauge field is encoded in the phases of these conditions.

For the evaluation of the VEV of the current density we have employed the direct summation over the complete set of fermionic modes. For the Bunch–Davies vacuum state, the corresponding mode functions are given by (2.13). Because of the compactification of the cosmic string along its axis, the quantum number corresponding to the z-direction becomes discrete and for the corresponding summation we use the Abel–Plana-type formula (3.4). As a consequence, the current density is decomposed in two parts: the first part corresponds to the geometry of a straight cosmic string in dS spacetime and the second one is induced by the compactification.

In the problem under consideration the VEVs of the charge density and the radial current vanish. For the VEV of the azimuthal current density we have considered first the part corresponding to the geometry without compactification, denoted by $\langle j^2 \rangle_s$. Two different integral representations for this part are provided, the expressions (3.11) and (3.18). The azimuthal current is an odd periodic function of the magnetic flux along the string axis with the period equal to the flux quantum. It depends on the radial and conformal time coordinates through the ratio r/η, which is the proper distance from the string measured in units of the dS curvature scale α. Near the string, the leading term in the asymptotic expansion of $\langle j^2 \rangle_s$ is independent of the mass and it behaves as the inverse forth power of the proper distance. In this limit the dominant contribution comes from the modes with small wavelengths and the effects of the curvature are small. For a massive field, the influence of the gravitational field on the vacuum current density is crucial at distances from the string larger than the curvature radius of the background spacetime. In this limit, the azimuthal current density exhibits a damping oscillatory behavior with the amplitude inversely proportional to the fourth power of the distance. Note that for the string in Minkowski spacetime and for a massive field, at large distances the current is exponentially suppressed.

The contribution to the azimuthal current density coming from the compactification of the string along its axis is given by (3.23) and the total current is given by (3.24). In the compactified geometry the azimuthal current is an odd periodic function of the magnetic flux along the string and an even periodic function of the flux enclosed by the z -axis. In both cases the period is equal to the flux quantum. Near the string the total azimuthal current is dominated by the part $\langle j^2 \rangle_s$. In this region the leading term in the expansion of topological part is given by (3.27). In the opposite limit of large distances from the string, the behavior of the azimuthal current density for a compactified cosmic string depends crucially on whether the parameter $\tilde{\beta}$ is zero or not. For $\tilde{\beta} \neq 0$ and $q \geqslant 2$, the leading term in the asymptotic expansion is given by (3.28) and the azimuthal current is suppressed by the factor $\mathrm{e}^{-4\pi\sigma_\beta r \sin(\pi/q)/L}$. For $q < 2$ the suppression is stronger, by the factor $\mathrm{e}^{-4\pi\sigma_\beta r/L}$. For $\tilde{\beta} = 0$, at large distances the azimuthal current exhibits a damping oscillatory behavior described by (3.29). The amplitude of the oscillations decay as $(r/\eta)^{-3}$ and in this case the topological part $\langle j^2 \rangle_c$ dominates in the total current density.

The VEV of the axial current density is given by the expression (4.4). The appearance of the nonzero axial current is a purely topological effect induced by the compactification of the string along its axis. The axial current density is an even periodic function of the magnetic flux along the string axis and an odd periodic function of the flux enclosed by the z-axis with the periods equal to the flux quantum. The modulus of the axial current, $|\langle j^3 \rangle|$, increases with increasing q for the values of the parameter a_0 close to $\pm 1/2$ and decreases for a_0 close to 0. In the absence of the planar angle deficit one has $q = 1$ and the general formula is reduced to (4.9). For general case of the parameter q, the corresponding asymptotic near the cosmic string is given by (4.11). At large distances from the string the effects of the planar angle deficit and of the magnetic flux on the axial current are small and to the leading order we recover the result for dS spacetime with a single compact dimension, described by (4.8). For small values of the proper length of the compact dimension, the axial current is dominated by the part corresponding to the current density in the geometry without the cosmic string and magnetic flux along the z-axis. In this limit, the leading term in the asymptotic expansion is given by (4.12) and the string-induced corrections to this leading term are exponentially small. For large values of the ratio L/η, the behavior of the axial current is described by (4.13). In this range the current density exhibits damping oscillations with power-law decaying amplitude as a function of the length of the compact dimension. This behavior is essentially different from the case of the string in Minkowski bulk with the exponentially suppressed axial current for large L.

Having in mind possible applications in models with compact extra dimensions, in section 5 we have generalized the expressions for the azimuthal and axial current densities in higher-dimensional background spacetimes with a single compact dimensions. In the way described above, more general geometries with an arbitrary number of toroidally compactified dimensions can be considered as well. In these models, the application of the summation formula (3.4) will lead to the recurrence relation for the current densities in models with spatial topologies $R^n \times (S^1)^q$ and $R^{n+1} \times (S^1)^{q-1}$ with $q = D - n$.

The results described above can be used, in particular, for the investigation of the effects induced by cosmic strings in the inflationary epoch. Though the strings produced before or during early stages of inflation are diluted by the expansion, cosmic strings can be continuously formed during inflation by coupling the symmetry breaking and inflaton fields [2] or by quantum-mechanical tunneling [64]. In the latter mechanism circular loops of string are created. The formation of cosmic strings during inflation can also be triggered by nonminimal coupling of the symmetry breaking field to the curvature of the background spacetime [65]. In hybrid models of inflation [66], that are naturally realized in the framework of supersymmetric theories, cosmic strings are produced at the the end of the inflationary phase. Another field of application corresponds to string-driven inflationary models with the cosmological expansion driven by the string energy [67]. Our results show that, in addition to be a source driving the inflation, the flux carrying strings induce electromagnetic effects having topological nature. These effects are present for both the straight and compactified cosmic string geometries. Moreover, they are induced by gauge field fluxes and survive in the absence of planar angle deficit as well.

The recent progress in models with large compact dimensions and large warp factors have shown that the fundamental strings can act as cosmic strings. The presence of compact dimensions is an inherent feature in the corresponding models. String compactifications with gauge field fluxes (for reviews see [68]) provide a framework in which the moduli fields are naturally stabilized. As we have demonstrated above, these fluxes are also sources of induced vacuum currents. In particular, in Kaluza–Klein type models the currents along compact dimensions generate magnetic field in the uncompactified subspace. The magnetic fields induced by the vacuum currents provide an additional channel for the interaction of cosmic strings with the environment.

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